

Unit :-2

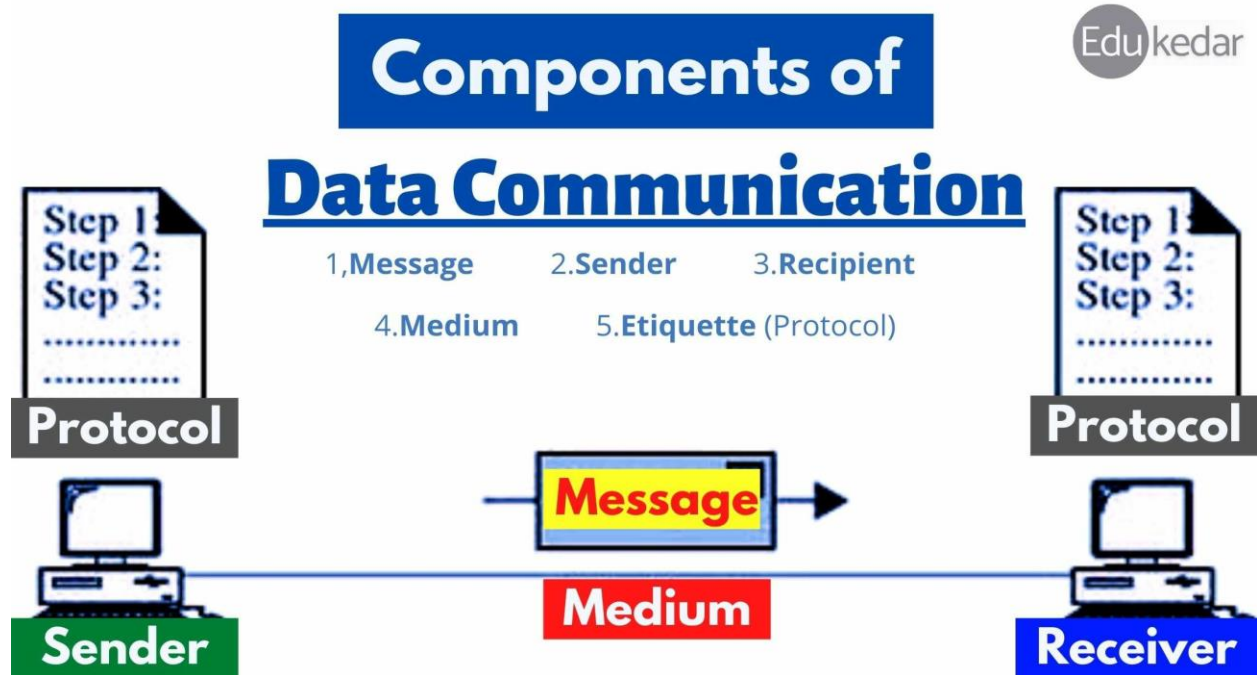
COMPUTER ARCHITECTURE

★ DATA COMMUNICATION :-

- Data Communication is the Process of sending or Receiving Digital data b/w Two or More Computers Via Transmission Medium such as a wire cable.

★ COMPONENT OF DATA COMMUNICATION ?

- Data Communication System has Five Components.
 - 1) Message
 - 2) Sender
 - 3) Receiver
 - 4) Transmission Medium
 - 5) Protocol



★ MODEM :-

- Modem stands For Modulator & Demodulator It is a network Device That is Placed B/W the Computer System & Telephone Line.
- It has Two Port modulator & demodulator Convert Digital Signal to Analog Signal Where as Demodulator Convert analog To Digital Signal.



[Fig. :- Modem]

Note :- It allow us to Computer To Connect Internet

★ NIC :-

- NIC stands for Network Interface Card, It Hardware Device Without which We can't Connect Computer to the Network/ Internet.

Types :-

- 1) Internal Network Card
- 2) External Network Card

1). Internal Network Card :-

- In this Cards, The Mother board has a Slot For the N/W Card Where it Can be Inserted.
- It Requires a Network Cable (Rj45) To Provide N/W access.



2). External Network Card :-

- In Desktop & Laptop that Don't have an Internal NIC. External NIC's are Used.

- It has Two Types : 1). Wireless

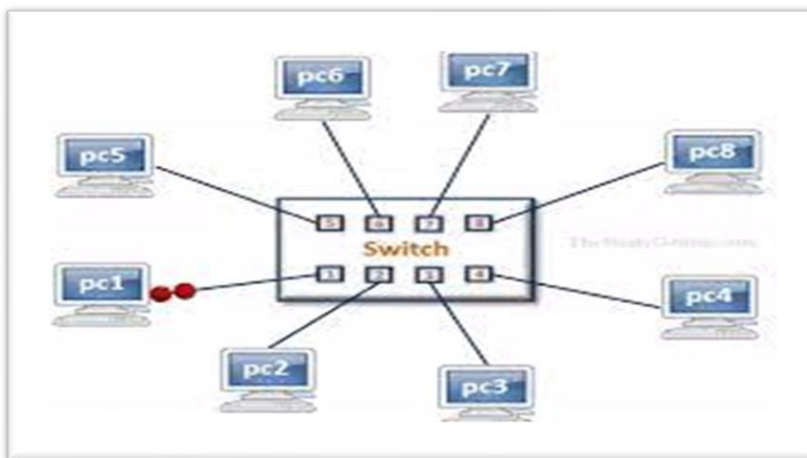


2). USB Cable

- Data is not Secured in this Cable.

★ WHAT IS SWITCH ? FULL EXPLANATION :-

- Switch :- Switch is a N/W device that Connects multiple Computers Together in the Network.
- It is mainly Used to send the Private message as well as there is no wasting of data.
- Switch Can easily Identify that which device is Connected with which part by using MAC address.
- That's why It delivered Message On Particular destination Machine.



[Fig :- Switch N/W device]

Note :- Switch is More Intelligent than HUB.

★ Advantage :-

1. It generally used to Unicast Message.
2. It Provides more Security than HUB.
3. Switch Support Full duplex data transmission mode.

4. It is Used to send the data Packet based on MAC address.
5. If a Node Fails There Will be no effect in the Entire Network.

★ **Dis advantage :-**

1). If Switch is failed then entire Network Will be Failed.

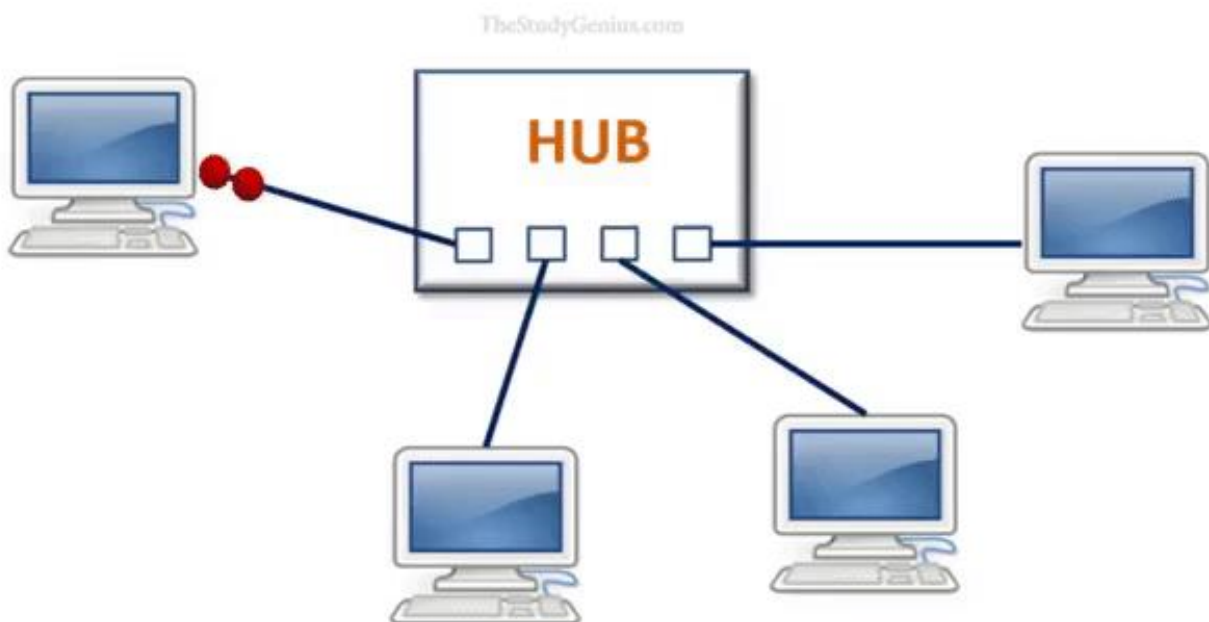
- It is More Expensive
- Difficult To setup

★ **WHAT IS HUB ?**

Hub Network Device :-

HUB :- Hub is Network device that is Used to Connect Multiple Computer in a Network.

- All the information send to the HUB is Automatically send to each Part To Every Device.
- A HUB is less Expensive less Intelligence & Less Complicated.
- HUB Generally Used to Connect Computers in LAN.
- Transmission mode of HUB is half Duplex.



[Fig :- HUB N/W Device]

★ Advantage :-

- The HUB Can Broadcast the message.
- It is Less Expensive that Anyone Can use it.
- Easy Installation.
- Robust

★ Dis Advantage :-

- If the HUB is Failed the Entire N/w will be Failed.
- We can't send Private/ Personal data Through hub.
- Hub can't support Full duplex Transmission Mode.

★ INPUT DEVICE :-

1. Joystick :- Joystick Is one type of gaming mouse.

- It is a Input device.
- This Device Is used For Playing games & using this device Children's easily learn the Computer.
- If you achieve any target this device also Warn you with Vibration effect.

2.Track Ball :-

- Track ball is look like small ball.

3.Light Pen :-

- Light Pen is used for making graphic on Screen.
- Mainly Light Pen is used in Online Classes.

4.Touch :-

- A Touch or Touch screen is the assembly Of both input (Touch panel)& output (display) Device.
- The touch Panel is Normally Layered on the top of on electronic Visual Display of an Information Processing System.
- A Users can give Input or control the Information Processing System through Simple or Multi-touch.

5.Voice Input System/ Voice Input Device :-

- A Device in Which Speech is Used to input data or System Commands Directly Into a System.
- Such Equipment involves the use of speech Recognition Processes & Can Replace or Supplement other input Devices.
- Some Voice input devices can recognize Spoken Words From a Predefined Vocabulary some have to be Trained For a Particular Speaker.
- Voice Device Used to enter Voice into Computer is called Voice Input Device.

★ **OUTPUT DEVICE :- (only Principles)**

1). Monitors :-

- Monitors Commonly Called as Visual Display Unit(VDU).are the main output device of a Computer.
- It Forms images From tiny dots Called Pixels that are Arranged in upon the number of pixels.
- There are Two Kinds of Viewing Screen Used For Monitors.
- 1). Cathode – Ray Tube (CRT)
- 2). Flat-panel Display.

★ **CRT Terminals :-**

Computer monitors Generate Output With the help Of Cathode Ray Tubes.

- This Type of Device is Called the CRT Monitor.
- It Is Larger Transportable Monitor Models.
- It is broad.
- It is can not used in small Portable Computer.

★ **LCD Monitor :-**

- LCD Monitor Display Units are Liquid Crystal Display Monitors.
- LCD and Plasma Screen Styles are Much Lighter than CRT monitor.
- LCD Models in Particular Require Less Power To Run.
- LCD is Costly.
- It occupies less space.

★ **LED Monitor :-**

- It Full Form is light emitting diode.

- LED have Better Responses Time than LCD.
- LED delivers good Picture Quality in Comparison to the LCD.
- LED Contrast ratio is high.
- It is Costly.
- LED Delivers better color accuracy in Comparison to the LCD.
- A Standard LED monitor User Light Emitting Diodes For Backlights.
- LED Resolution High.

★ **Printers :-**

- Printer is an output device. which is Used to print Information on Paper.
- There are two Types of Printers.

1).Impact Printers

2).Non Impact Printers

1).Impact Printers :-

Impact printers Print the Characters By Striking Them on the Ribbon, Which is then Pressed on the Paper.

Characteristics Of Impact Printers :-

- Very low Consumable Costs
- Very Noisy
- Useful For bulk Printing due to low cost.
- There is Physical Contact with the Paper to Produce an image.

❖ **There Printers are two types :-**

1). Character Printers.

2). Line Printers.

1). Character Printers :- character Printers are the Printers Which Print one character at a time.

These are Further Divided into two Types :-

1). Dot Matrix Printer (DMP)

2). Daisy Wheel

1). Dot Matrix Printer :- One of the Most Dot matrix Printer.

- These Printer are Popular because of their ease of Printing & Economical Price.
- Each character Printed is in the Form of Pattern of dots and head Consists Of a Matrix Of Pins of size.(5*7,7*9,9*7 or 9*9) Which come out to Form a character which is Why it is Called Dot matrix Printer.

★ Advantages :-

- Inexpensive
- Widely Used
- Other Language Characters can be Printer.

★ Dis Advantages :-

- Slow Speed
- Poor Quality

★ 2). Daisy Wheel :-

- Head is Lying on a Wheel & Pins Corresponding To Characters are Like Petals of Daisy (Flower) Which Is Why it is called Daisy wheel Printer.
- These Printers are generally Used For Word Processing in Offices that require a few Letters To be sent here & there Will Very Nice Quality.

★ Advantages :-

- More Reliable the DMP.
- Better Quality.
- Fonts of Character can be easily Changed.

★ Dis Advantages :-

- Slower Than DMP.
- Noisy
- More Expensive than DMP.

★ Line Printers :- Line Printers are the Printers Which Print One Line at a time.

- There are Of Two Types.
- 1). Drum Printer.
 - 2). Chain Printer.

1). Drum Printer :-

- This Printer is like a Drum in shape that's why it is called Drum Printer.
- The Surface of the drum is Divided into a number of tracks.

★Advantages :-

- Very high speed

★Dis Advantages :-

- Very Expensive.
- Characters Fonts can not be changed.

★Chain Printer :- In this Printer a Chain of Character sets is Used.

- A Standard Character set may have 48,64, or 96 characters.

★Advantages :-

- Character Fonts can Easily be changed.
- Different languages can be used With the Same Printer.

★Dis Advantages :-

- Noisy

★Non-Impact Printers :-

- Non Impact Printers Print the Characters Without Using The Ribbon.
- These Printers Print a Complete Page at a Time they are also Called as Page Printers.

These Printers are of Two Types :-

1). Laser Printers

2). Inkjet Printers

Characteristics Of Non-impact Printers :-

- Faster than Impact Printer
- They are not Noisy.
- High Quality
- Support many Fonts & Different character Size.

1). Laser Printers :- There are Non impact page Printers.

- They use laser Lights to Produce the dots needed to Form the Characters to be Printed on a page.



Advantages :-

- Very high Speed
- Very High Quality Output
- Good Graphics Quality
- Supports many Fonts & Different character Size.



Dis Advantages :-

- Expensive
- Cannot be used to Produce Multiple Copies of a Document in a Single Printing.

2). Inkjet Printer :-

- Inkjet Printers are Non Impact.character Printers based on a Relatively.
- They Print Characters by Spraying small Drops of ink on to Paper.
- Inkjet Printers Produce High Quality Output With Presentable Features.



Advantages :-

- High Quality Printing More Reliable.



Dis Advantages :-

- Slow as Compared to Laser Printer.
- Expensive as The Cast Per page is High.



Plotter :-

- This Device is similar to a Printer which is used for graphical Output like pie Charts, bar charts & Graphs.
- Engineers & architecture Use this device For Designing.
- Plotters Produce High Quality Drawings and Graphs.
- It is a special Purpose Device.
- There are Two Types Of Plotters :-

1). Drum Plotter

2).Flat bed plotter

★ **STORAGE DEVICE :-**

1). Magnetic Disk :-

- A Magnetic Disk is a Storage Device that uses a magnetization Process to Write Rewrite and Access data.
- It is covered with a magnetic coating and stores data in the Form of tracks spots and sectors.
- Hard disk & Floppy disks are common Examples of magnetic disks.
- A magnetic disk drive Consists of a number of Platters(disk) Coated in a Magnetic Material.

2).Floppy Disk :-

- A Portable Storage medium Consisting of a Square plastic Case Containing a thin disk coated with Magnetic material that is inserted into a Drive to read & Write data: Popular From The 1970s Through the 1990s.

3).Zip Disk :-

- The Zip Drive is a Removable Floppy disk storage system that was introduced by Lomega in late 1994.

★ **WHAT IS HARD DISK DRIVE ?**

- Hard disk is an Expensive Storage Device.
- Because this drive Contains Disc as Well as Drive.
- It Storing Capacity is Highest.
- This Device Works on a Direct Storage Method.
- It is a Sealed aluminium box.
- On a Hard disk. Data is stored in thin bonds.
- A drive head can read or Write a Circular ring or band Called a Track.
- There can be more than a Thousand tracks on a 3.5 inch hard disk.
- Section of each track are Called Sectors.
- A Sector is the smallest Physical Storage Unit on a Disk Which store 512 bytes in Size.
- A Cylinder is Formed While all Drive heads are in the same Position on the disk.

- The tracks stacked on top of each other Form a Cylinder.

★ **SEMICONDUCTOR MEMORY :-**

- This memory uses transistor and resisters For Storage.
Ex. Rom, PROM, &RAM
- The basic element of a semiconductor Memory is the Memory cell.
- Semiconductor memory devided in to Two Types.
 - 1) Semiconductor Main Memory
 - 2) Basic Memory Cell.

★ **FLASH MEMORY :-**

- Flash Memory is a kind of memory that retains data in the absence of Power Supply.
EX:- Semiconductor memory, CD,SD CARD etc.

★ **OPTICAL MEMORY :-**

- There Device Store more than Floppy but less than hard disk.
- They are much Reliable as Compare to Floppy disk.
- It is the Necessity To store large Volume of data that has led to the development Of optical disk storage medium.
- Optical memory devided in to following Types.
CD, CD-ROM, CD-RAM ,DVD ,DVD- ROM, DVD-RAM
- CD-ROM :- CD Rom disks are made of reflective metals.
- CD-ROM is Written During the Process of Manufacturing by high Power laser beam.
- In CD ROM Storage density is Very High.
- Storage cost is very low and access time is relatively fast.
- CD ROM can be read only, We can nit Write or make Changes into the data Contained in it.

DVD :-

- DVD stands for Digital Versatile Disk or Digital Video Disk.
- It is Possible to have 4.7 GB.
- GB ordinary data or mix DVD-Video & data on a Recordable DVD that can be Played by Most Computer DVD-ROMs.

★ **CACHE MEMORY :-**

1).Physical Memory :-

- Placed in a slot on the Computer Motherboard.
- Allocates information in a “First in, Last Out” Process.
- Faster Accessing Rate.
- Physical Memory is limited to the size of the RAM Chips.
- Data can be added or Removed.
- Can Directly access CPU.

2).VIRTUAL MEMORY :-

- Virtual Memory is Stored on the hard disk.
- Virtual Memory Uses a Process called Paging.
- Slower accessing Rate.
- Virtual Memory is limited by the size of The hard disk.
- Data can not be added or removed again and again.
- Can not access CPU Directly.